

Chunfan Yao

UNDERGRADUATE RESEARCHER · LLM EVALUATION, VERIFIERS AND OPTIMIZATION

Berkeley, CA (Exchange) / Guangzhou, China

☎ (+86) 186-2061-1072 | ✉ cfyao_berk@berkeley.edu | 🏎️ Ferrari-72

Summary

Undergraduate AI researcher at HKUST(GZ), visiting UC Berkeley in Fall 2026. I build evaluation and verification infrastructure for coding and research agents—including runners, certificate-based verifiers, independent review, and formal-verification gates—and graph-search methods for Rydberg-atom optimization.

Research Experience

Prof. Jinguo Liu's Group, HKUST(GZ)

Guangzhou, China

LLM BUG-FINDING BENCHMARK (Evaluation & Verification)

2025 – 2026

- Developed the initial SWE-style and mini-SWE-agent runners for [problem-reductions-benchmark](#), which evaluates LLM agents on subtle logical bugs across a combinatorial-reduction repository.
- Implemented the `pred`-based verifier and certificate pipeline, including checks for solve mismatch, extraction unsoundness, order reversal, incomplete reductions, and novelty.
- Added an `AgentRunner` interface, concurrent scheduler, trajectory archiving, and 99 automated tests; defined bugs/Ktok and bugs/dollar metrics with reproducible run outputs and leaderboard infrastructure.

AutoResearch for Reduction-Rule Discovery

Guangzhou, China

RESEARCH WORKFLOW DESIGN (Trustworthy Research Agents)

2025 – Present

- Designed a verify-first agentic pipeline with fixed polynomial-time F/G contracts and ≥ 100 independently labeled instances before generation; preserved counterexamples across propose–test–diagnose rounds and separated no-shared-code verification, cold-context review, and optional Lean formalization into explicit assurance gates.

Quantum Computing & Graph Optimization Lab — Prof. Jinguo Liu, HKUST(GZ)

Guangzhou, China

UNDERGRADUATE RESEARCH ASSISTANT (Co-First Author)

2025 – Present

- Co-authoring a manuscript in preparation for submission to *PRX Quantum* on unit-weight CROSS and CROSS+EDGE gadgets for triangular Rydberg lattices, enabling universal MIS encoding under a single uniform detuning.
- Proposed the vertex-split decomposition that unblocked the construction by splitting a high-degree pin into two lower-degree vertices while preserving the reduced tensor.
- Implemented reduced- α tensor and gadget-equivalence checks plus an unweighted-search API in [GadgetSearch.jl](#), supporting certification and search over 5.6B candidates ($n \leq 15$).
- Diagnosed failed RL/search approaches and helped redirect the project toward constructive graph decomposition and exhaustive certification.

Industry Experience

Hikvision (Hangzhou Headquarters)

Hangzhou, China

SOLUTIONS ENGINEER INTERN

Jun. 2026 – Jul. 2026

- Supported pre-sales solution design for smart-city AI deployments by translating client requirements into system architectures, integration documents, and technical demonstrations of production video-analytics pipelines.

Education

The Hong Kong University of Science and Technology (Guangzhou)

Guangzhou, China

B.ENG. IN ARTIFICIAL INTELLIGENCE

Sep. 2024 – Present

University of California, Berkeley

Berkeley, CA

VISITING UNDERGRADUATE (EXCHANGE)

Aug. 2026 – Dec. 2026

Technical Skills

Languages	Python, Julia, Git, Linux/CLI, $\text{\LaTeX}$; reading familiarity with Rust
Agent/Eval	LLM evaluation harnesses, executable verifiers, benchmark design, RAG/context retrieval
Methods	Combinatorial optimization, graph algorithms, PyTorch, Transformers, experiment automation